

STA623 - Bayesian Data Analysis

Practical 1: Bayesian reasoning: prior, likelihood, posterior (with solutions)

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Goals. Use Bayes' rule numerically; compute posteriors by grid approximation for one-parameter models; derive a conjugate posterior and check grid results against it; see how prior and sample size interact.

Exercise 1: combining two diagnostic tests

A new diagnostic test for a disease D has sensitivity $P(T \mid D) = 0.99$ and specificity $P(\bar{T} \mid \bar{D}) = 0.90$. An existing test has sensitivity 0.80 and specificity 0.95. Given disease status, the two tests are independent.

- (a) With prevalence 10%, compute $P(D \mid T)$ for each test, and $P(D \mid \text{both positive})$. *Hint: define a compound test that is positive only if both tests are positive.*
- (b) Write an R function `ppv(prev, sens, spec)` and plot $P(D \mid T+)$ against prevalence (0.1% to 50%) for the new test, the old test and the compound test on one graph. At which prevalence does the new test's positive predictive value reach 50%?
- (c) Sequential testing: a patient tests positive on the new test. Use the posterior from that result as the *prior* for the old test. Show that this gives the same answer as the compound test in (a). Why?

Solution

- (a) By Bayes' rule, with $P(D) = 0.1$ and $P(\bar{D}) = 0.9$, for the new test:

$$P(D \mid T) = \frac{P(T \mid D)P(D)}{P(T \mid D)P(D) + P(T \mid \bar{D})P(\bar{D})} = \frac{0.99 \times 0.1}{0.99 \times 0.1 + (1 - 0.90) \times 0.9} = \frac{0.099}{0.189} = 0.5238.$$

For the existing test:

$$P(D \mid T_{\text{old}}) = \frac{0.80 \times 0.1}{0.80 \times 0.1 + (1 - 0.95) \times 0.9} = \frac{0.080}{0.125} = 0.6400.$$

The compound test $U = T \cap T_{\text{old}}$ has, by conditional independence of the two tests given disease status,

$$P(U \mid D) = 0.99 \times 0.80 = 0.792, \quad P(U \mid \bar{D}) = (1 - 0.90)(1 - 0.95) = 0.005,$$

so

$$P(D \mid U) = \frac{0.792 \times 0.1}{0.792 \times 0.1 + 0.005 \times 0.9} = \frac{0.0792}{0.0837} = 0.9462.$$

The new test, despite its much higher sensitivity, has a *lower* positive predictive value than the old one: with 90% of people disease-free, its lower specificity produces many false positives. Requiring both tests to be positive raises the probability of disease to about 95%.

(b)

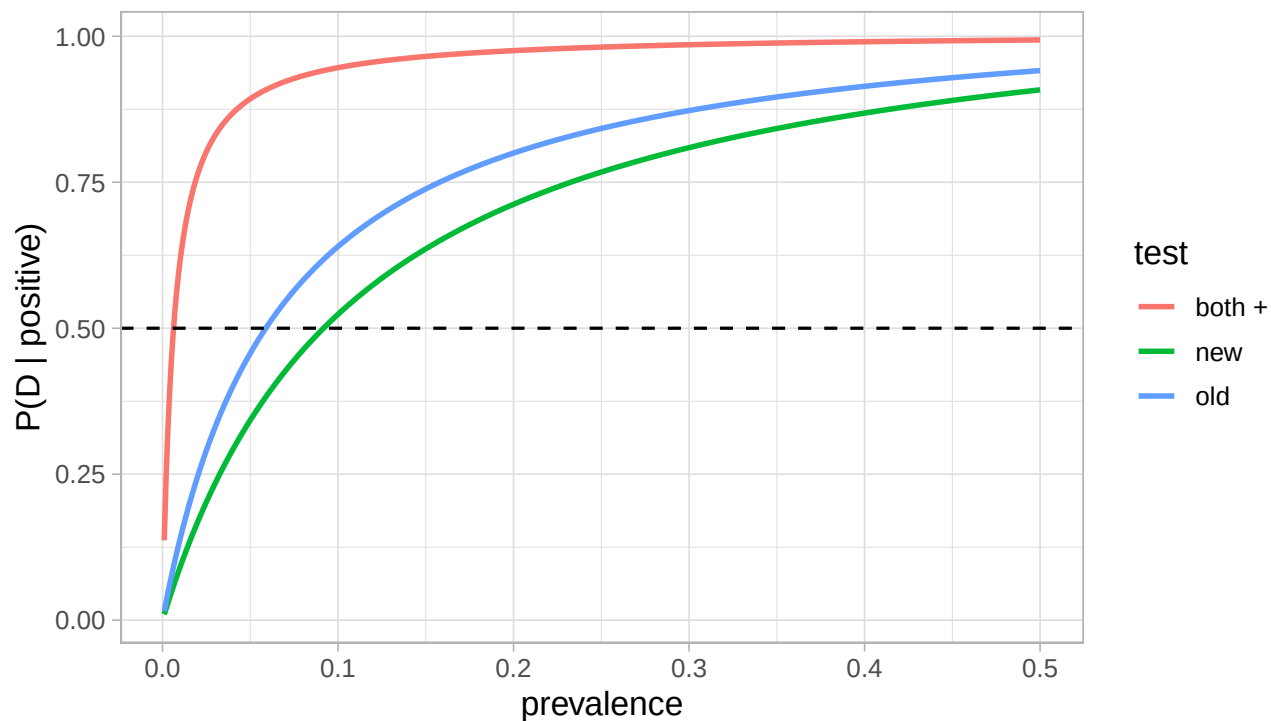
```
ppv <- function(prev, sens, spec) {
  sens * prev / (sens * prev + (1 - spec) * (1 - prev))
}

tests <- tribble(
  ~test,      ~sens,      ~spec,
  "new",      0.99,      0.90,
  "old",      0.80,      0.95,
  "both +",   0.99 * 0.80, 1 - (1 - 0.90) * (1 - 0.95))

tests |> mutate(`P(D | positive), prevalence 10%` = ppv(0.10, sens, spec)) |>
  kable(digits = 4) # check: matches the hand calculations in (a)
```

test	sens	spec	P(D positive), prevalence 10%
new	0.990	0.900	0.5238
old	0.800	0.950	0.6400
both +	0.792	0.995	0.9462

```
expand_grid(prev = seq(0.001, 0.5, length = 500), tests) |>
  mutate(ppv = ppv(prev, sens, spec)) |>
  ggplot(aes(prev, ppv, colour = test)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 0.5, linetype = 2) +
  labs(x = "prevalence", y = "P(D | positive)")
```



```
uniroot(function(p) ppv(p, 0.99, 0.90) - 0.5, c(0.001, 0.5))$root
```

```
[1] 0.09174252
```

The new test's PPV reaches 50% at a prevalence of about 9.2%. Below that, a positive result from the new test is more likely to be a false positive than a true one.

(c)

```
post_after_new <- ppv(0.10, 0.99, 0.90)
ppv(post_after_new, 0.80, 0.95)
```

```
[1] 0.9462366
```

Identical to the compound test. Bayesian updating is coherent: updating on two conditionally independent pieces of evidence one after the other gives the same posterior as updating on both at once, because $P(D | T, T_{\text{old}}) \propto P(T_{\text{old}} | D) P(T | D) P(D)$ whichever way we group the factors.

Exercise 2: grid approximation for a prevalence

A school survey tests $n = 40$ children for *Schistosoma haematobium*; $k = 11$ are positive. Earlier surveys in the district suggest a prevalence around 30%, and you encode this as a $\text{Beta}(3, 7)$ prior.

- (a) **Exact posterior.** Assuming $k \mid \pi \sim \text{Bin}(n, \pi)$ and the prior $\pi \sim \text{Beta}(a, b)$, derive the posterior distribution $p(\pi \mid k)$ analytically and state its parameters for these data. Then use R (`qbeta`, `pbeta`) to compute the exact posterior mean, median, 95% equal-tailed credible interval and $P(\pi > 0.35 \mid k)$.
- (b) Compute the posterior of the prevalence π on a grid of 1001 points. Plot prior, (scaled) likelihood and posterior.
- (c) Draw 10^5 samples from the grid posterior and use them to estimate the posterior mean, median, a 95% credible interval and $P(\pi > 0.35 \mid k)$.
- (d) Compare with the exact posterior summaries from (a).
- (e) Repeat (b) using a grid of only 11 points. What goes wrong?

Solution

- (a) The binomial likelihood and the Beta prior are

$$p(k \mid \pi) = \binom{n}{k} \pi^k (1 - \pi)^{n-k}, \quad p(\pi) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \pi^{a-1} (1 - \pi)^{b-1}.$$

Keeping only the factors that depend on π :

$$p(\pi \mid k) \propto \pi^k (1 - \pi)^{n-k} \cdot \pi^{a-1} (1 - \pi)^{b-1} = \pi^{a+k-1} (1 - \pi)^{b+n-k-1},$$

which is the kernel of a Beta density. Hence

$$\pi \mid k \sim \text{Beta}(a + k, b + n - k) = \text{Beta}(3 + 11, 7 + 40 - 11) = \text{Beta}(14, 36).$$

The posterior mean is $14/50 = 0.28$.

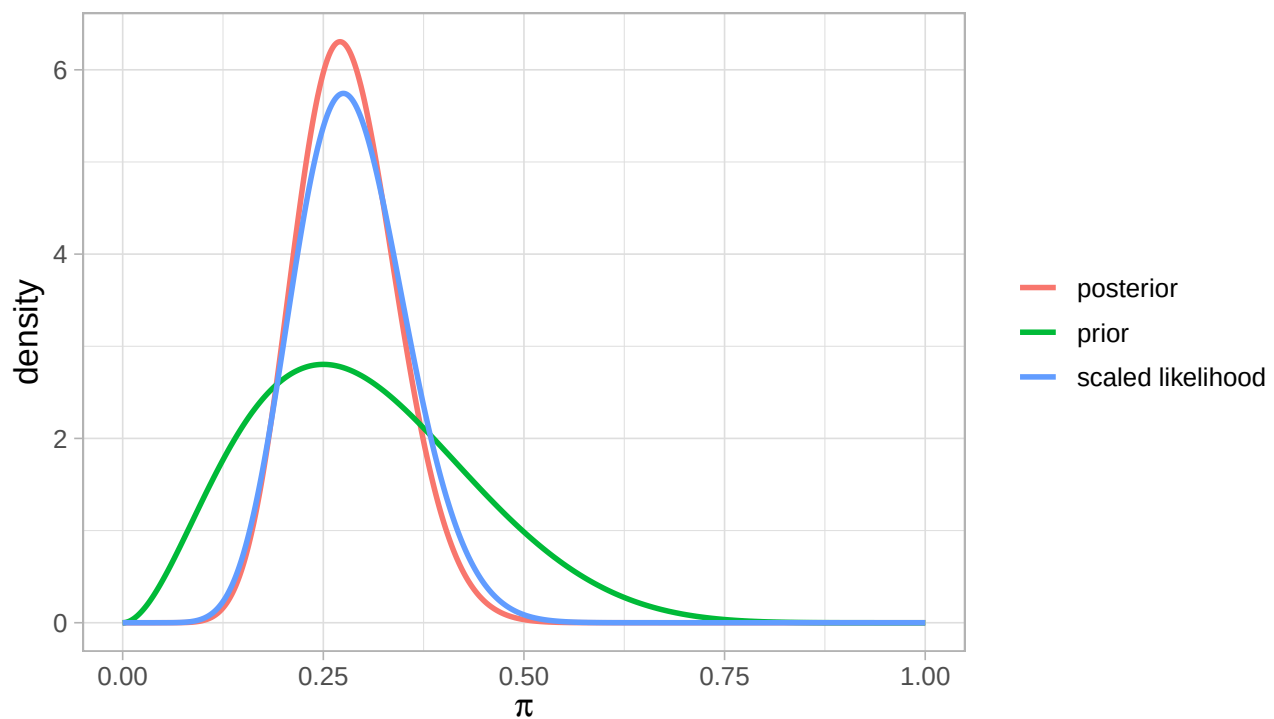
```
n <- 40; k <- 11; a <- 3; b <- 7
a_post <- a + k; b_post <- b + n - k
exact <- c(mean = a_post / (a_post + b_post),
           median = qbeta(0.5, a_post, b_post),
           setNames(qbeta(c(0.025, 0.975), a_post, b_post), c("2.5%", "97.5%")),
           `P(pi>0.35)` = 1 - pbeta(0.35, a_post, b_post))
round(exact, 4)
```

mean	median	2.5%	97.5%	P(pi>0.35)
0.2800	0.2770	0.1658	0.4108	0.1363

(b)

```
h <- 0.001
grid <- tibble(pi = seq(0, 1, by = h)) |>
  mutate(prior = dbeta(pi, a, b),
         lik = dbinom(k, n, pi),
         post = prior * lik / sum(prior * lik * h))

grid |>
  mutate(`scaled likelihood` = lik / sum(lik * h)) |>
  select(pi, prior, `scaled likelihood`, posterior = post) |>
  pivot_longer(-pi) |>
  ggplot(aes(pi, value, colour = name)) + geom_line(linewidth = 1) +
  labs(x = expression(pi), y = "density", colour = NULL)
```



(c) and (d)

```
draws <- sample(grid$pi, 1e5, replace = TRUE, prob = grid$post)

rbind(grid_sampling = c(mean = mean(draws), median = median(draws),
                        quantile(draws, c(0.025, 0.975)), `P(pi>0.35)` =
                        ↪ mean(draws > 0.35)),
      exact = exact) |>
kable(digits = 4)
```

	mean	median	2.5%	97.5%	P(pi>0.35)
grid_sampling	0.28	0.277	0.1650	0.4100	0.1376
exact	0.28	0.277	0.1658	0.4108	0.1363

The two agree to two or three decimal places; the remaining differences are Monte Carlo error and grid discretisation.

(e)

```
coarse <- tibble(pi = seq(0, 1, by = 0.1)) |>
  mutate(post = dbeta(pi, a, b) * dbinom(k, n, pi), post = post / sum(post))
coarse_draws <- sample(coarse$pi, 1e5, replace = TRUE, prob = coarse$post)
c(mean = mean(coarse_draws), quantile(coarse_draws, c(0.025, 0.975)),
  ↪ `P(pi>0.35)` = mean(coarse_draws > 0.35))
```

mean	2.5%	97.5%	P(pi>0.35)
0.279734	0.200000	0.400000	0.112460

The mean is still roughly right but the interval and tail probability can only take values on the coarse grid: $P(\pi > 0.35)$ is really $P(\pi \geq 0.4)$ here. Grids must be fine enough to resolve the posterior, which in more dimensions quickly becomes infeasible.

Exercise 3: a discrete prior

A bag contains six coins: three with $P(\text{heads}) = 0.8$, two with 0.3 and one fair coin. One coin is drawn at random and tossed $n = 6$ times, giving $k = 2$ heads.

- (a) Compute the posterior probability of each coin type. (This is a grid with three points.)
- (b) The same coin is tossed another 6 times, giving 5 heads. Update your posterior. Then check that analysing all 12 tosses at once gives the same result.
- (c) Can this analysis ever conclude that $\pi = 0.6$? Comment.

Solution

```
coins <- tibble(pi = c(0.3, 0.5, 0.8), prior = c(2, 1, 3) / 6) |>
  mutate(lik1 = dbinom(2, 6, pi),
         post1 = prior * lik1 / sum(prior * lik1),
         lik2 = dbinom(5, 6, pi),
         post2 = post1 * lik2 / sum(post1 * lik2),
         post_all = prior * dbinom(7, 12, pi) / sum(prior * dbinom(7, 12, pi)))
kable(coins, digits = 4)
```

pi	prior	lik1	post1	lik2	post2	post_all
0.3	0.3333	0.3241	0.6980	0.0102	0.1417	0.1417
0.5	0.1667	0.2344	0.2524	0.0937	0.4704	0.4704
0.8	0.5000	0.0154	0.0496	0.3932	0.3879	0.3879

- (a) After 2 heads in 6 tosses, the coin with $\pi = 0.3$ is most probable (about 0.70).
- (b) Sequential updating (`post2`) and the single analysis of 7 heads in 12 tosses (`post_all`) agree exactly (the binomial coefficients cancel on normalisation).
- (c) No: values with zero prior probability always have zero posterior probability. That is appropriate here because we *know* the coins in the bag, but in general a prior should never exclude physically possible values.

Exercise 4: grid approximation for a Poisson rate, conjugate and non-conjugate

Number of malaria episodes in one season for 18 children:

```
y <- c(1, 3, 2, 3, 0, 2, 6, 4, 4, 1, 1, 3, 2, 3, 1, 1, 3, 0)
```

Model: $Y_i \mid \lambda \sim_{\text{iid}} \text{Pois}(\lambda)$, with prior $\lambda \sim \text{Gamma}(a, b)$, $a = 5$, $b = 2$ (shape a , rate b).

- (a) **Exact posterior.** Derive the posterior distribution $p(\lambda \mid \mathbf{y})$ analytically. *Hint: write down the likelihood and the prior, keep only the factors that depend on λ , and recognise the kernel of a known distribution.* State its parameters for these data, and use R to compute the exact posterior mean and 95% equal-tailed credible interval. Through which summary of the data does $\mathbf{y}$ enter the posterior?
- (b) **Grid approximation.** Compute the posterior of λ on a grid over $(0, 6)$, working on the **log scale** (compute log-likelihood + log-prior, subtract the maximum, exponentiate, then normalise by dividing by the sum over the grid). Plot it together with your exact posterior from (a), and compare the posterior mean and 95% credible interval obtained from the grid with the exact values.
- (c) Now use a prior that is *not* conjugate: a half-normal, $\lambda \sim \mathcal{N}^+(0, 3^2)$ (use `2 * dnorm(lambda, 0, 3)`). Can you still derive the posterior in closed form? Compute the grid posterior, its mean and 95% credible interval. Does the change of prior matter?
- (d) Why is it important to work on the log scale? Try computing `prod(dpois(rep(y, 50), 2))` (i.e. 50 times as much data).

Solution

- (a) The likelihood of n independent Poisson observations is

$$p(\mathbf{y} \mid \lambda) = \prod_{i=1}^n \frac{\lambda^{y_i} e^{-\lambda}}{y_i!} = \frac{1}{\prod_i y_i!} \lambda^{\sum_i y_i} e^{-n\lambda},$$

and the prior density is $p(\lambda) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda}$. Dropping all factors that do not depend on λ :

$$p(\lambda \mid \mathbf{y}) \propto p(\mathbf{y} \mid \lambda) p(\lambda) \propto \lambda^{\sum_i y_i} e^{-n\lambda} \cdot \lambda^{a-1} e^{-b\lambda} = \lambda^{a+\sum_i y_i-1} e^{-(b+n)\lambda}.$$

This is the kernel of a Gamma density, so

$$\lambda \mid \mathbf{y} \sim \text{Gamma}\left(a + \sum_i y_i, b + n\right).$$

No integral is needed: the normalising constant must be the one that makes this a Gamma density. The Gamma prior is conjugate for the Poisson sampling model. The data enter only through n and $\sum_i y_i$, the sufficient statistic.

Here $n = 18$ and $\sum_i y_i = 40$, so the posterior is $\text{Gamma}(45, 20)$, with mean $45/20 = 2.25$:

```
a <- 5; b <- 2
a_post <- a + sum(y); b_post <- b + length(y)
c(shape = a_post, rate = b_post)
```

```
shape rate
    45   20
```

```
exact <- c(mean = a_post / b_post,
           setNames(qgamma(c(0.025, 0.975), a_post, b_post), c("2.5%", "97.5%")))
exact
```

```
      mean      2.5%      97.5%
2.250000 1.641165 2.953397
```

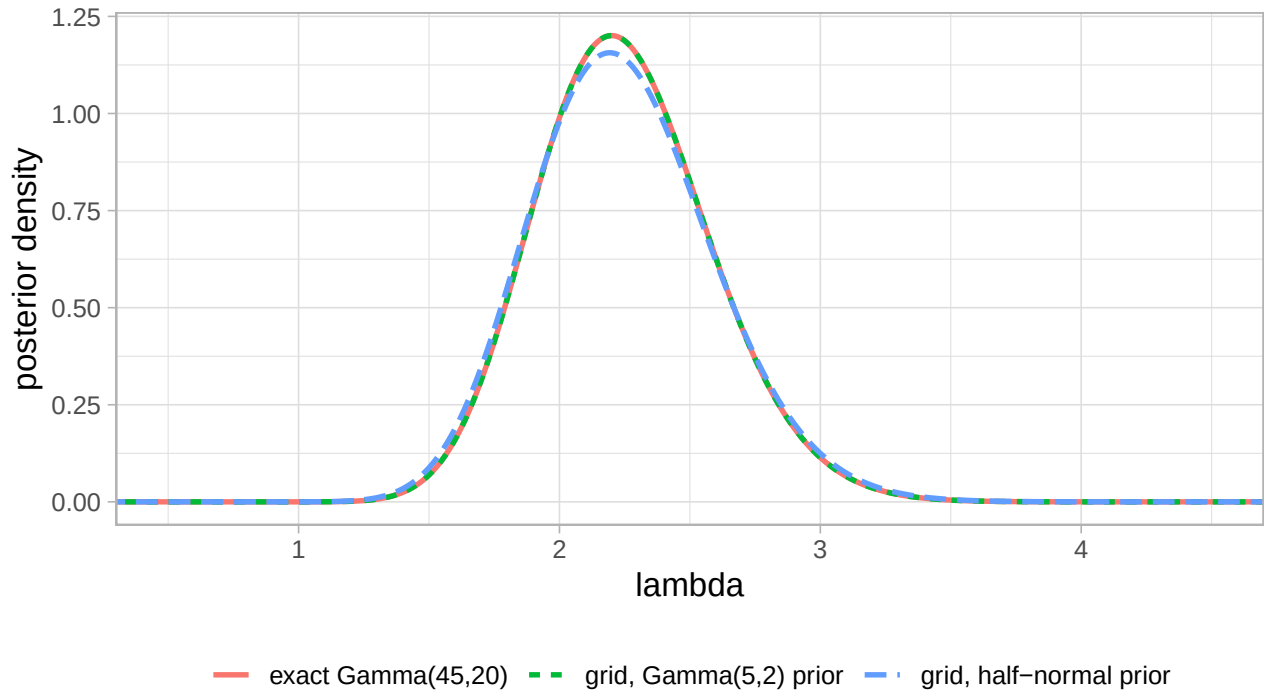
(b)–(c)

```
log_lik <- function(lambda) sapply(lambda, function(l) sum(dpois(y, l, log =
  ↪ TRUE)))
lam <- seq(0.001, 6, length = 3000); h <- diff(lam)[1]

grid_post <- function(log_prior) {
  lp <- log_lik(lam) + log_prior
  p <- exp(lp - max(lp))
  p / sum(p * h)
}

post_gamma <- grid_post(dgamma(lam, a, b, log = TRUE))
post_hn    <- grid_post(log(2) + dnorm(lam, 0, 3, log = TRUE))

tibble(lambda = lam, `grid, Gamma(5,2) prior` = post_gamma,
       `exact Gamma(45,20)` = dgamma(lam, a_post, b_post),
       `grid, half-normal prior` = post_hn) |>
  pivot_longer(-lambda) |>
  ggplot(aes(lambda, value, colour = name, linetype = name)) + geom_line(linewidth
  ↪ = 1) +
  coord_cartesian(xlim = c(0.5, 4.5)) +
  labs(y = "posterior density", colour = NULL, linetype = NULL) +
  theme(legend.position = "bottom")
```



```
summ <- function(p) {
  d <- sample(lam, 1e5, replace = TRUE, prob = p)
  c(mean = mean(d), quantile(d, c(0.025, 0.975)))
}
rbind(exact_gamma_posterior = exact,
      grid_gamma_prior      = summ(post_gamma),
      grid_half_normal      = summ(post_hn)) |>
kable(digits = 3)
```

	mean	2.5%	97.5%
exact_gamma_posterior	2.250	1.641	2.953
grid_gamma_prior	2.251	1.639	2.957
grid_half_normal	2.245	1.617	2.977

- (b) The grid posterior and the exact Gamma(45, 20) curve are indistinguishable, and the grid mean and interval agree with the exact values to about two decimal places. The small remaining differences are Monte Carlo error from sampling the grid.
- (c) No closed form: the half-normal prior is not conjugate, so $\lambda^{\sum y_i} e^{-n\lambda} e^{-\lambda^2/18}$ is not the kernel of a standard distribution. The grid method does not care. The half-normal prior gives an almost identical posterior mean and a very slightly wider interval (it is less concentrated than Gamma(5, 2)): 18 observations already carry most of the information.
- (d)

```
prod(dpois(rep(y, 50), 2))      # underflows to exactly 0
```

```
[1] 0
```

```
sum(dpois(rep(y, 50), 2, log = TRUE))
```

```
[1] -1612.386
```

The product of 900 probabilities is smaller than the smallest representable double and becomes 0, so every grid point would get posterior 0 and normalisation would fail (0/0). Sums of logs are perfectly well behaved.

Exercise 5: how much does the prior matter?

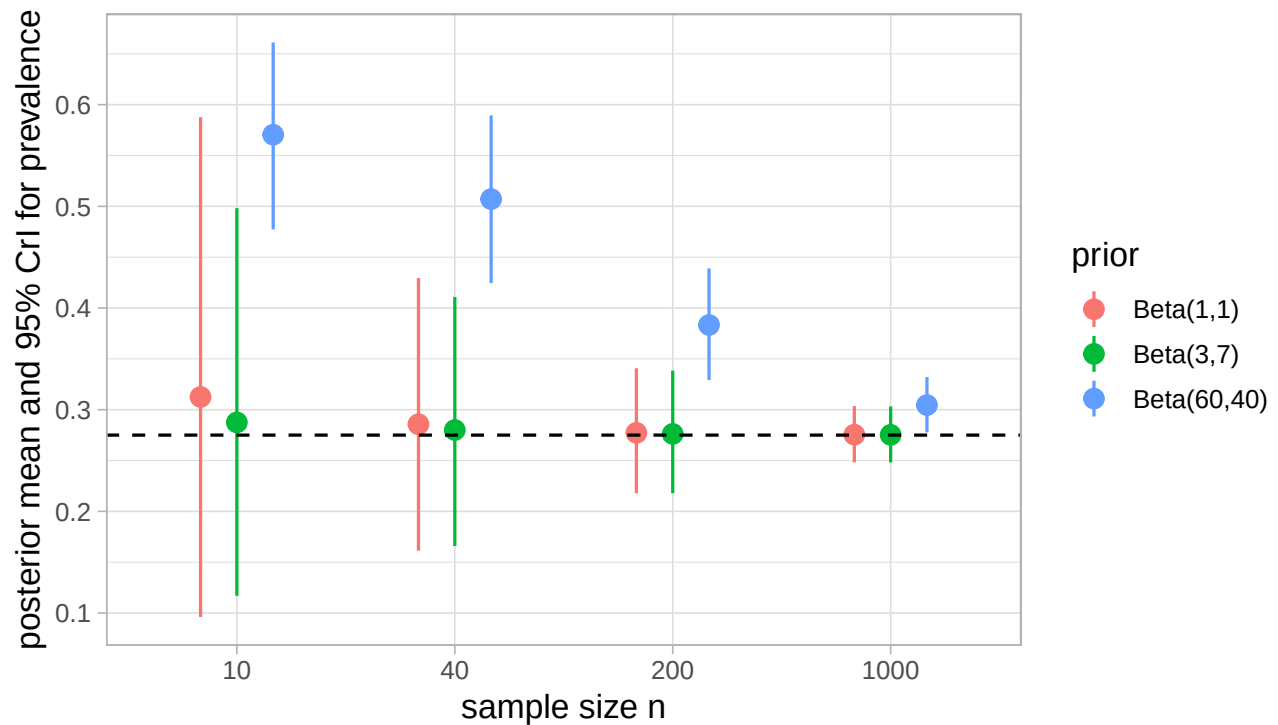
For the prevalence setting of Exercise 2, suppose the observed proportion stays at 27.5% while the sample size is $n = 10, 40, 200, 1000$.

- (a) For each n , compute the posterior mean and 95% credible interval under three priors: Beta(1, 1), Beta(3, 7) and a strongly informative but *wrong* Beta(60, 40) (prior mean 60%).
- (b) Plot the results. Describe when the prior matters, and how the “prior sample size” $a + b$ helps you predict this.

Solution

```
res <- expand_grid(n = c(10, 40, 200, 1000),
                  prior = c("Beta(1,1)", "Beta(3,7)", "Beta(60,40)")) |>
  mutate(k = 0.275 * n,
         a = c(1, 3, 60)[match(prior, c("Beta(1,1)", "Beta(3,7)",
          ↪ "Beta(60,40)"))],
         b = c(1, 7, 40)[match(prior, c("Beta(1,1)", "Beta(3,7)",
          ↪ "Beta(60,40)"))],
         mean = (a + k) / (a + b + n),
         lo = qbeta(0.025, a + k, b + n - k),
         hi = qbeta(0.975, a + k, b + n - k))

ggplot(res, aes(factor(n), mean, colour = prior)) +
  geom_pointrange(aes(ymin = lo, ymax = hi), position = position_dodge(width =
    ↪ 0.5)) +
  geom_hline(yintercept = 0.275, linetype = 2) +
  labs(x = "sample size n", y = "posterior mean and 95% CrI for prevalence",
    ↪ colour = "prior")
```



With $n = 10$ the three priors give very different answers; by $n = 1000$ they almost coincide. The weight of the prior in the posterior mean is $(a + b)/(a + b + n)$: the Beta(60,40) prior is worth 100 observations, so it dominates at $n = 10$ and $n = 40$, still has about a third of the weight at $n = 200$, and fades only at $n = 1000$. A strong prior that conflicts with the data is dangerous in small studies; Lecture 5's posterior predictive checks give a way to detect such conflicts.

Exercise 6

Suppose $\Pi \sim \text{Beta}(a, b)$ and $Y \mid \pi \sim \text{Bin}(n, \pi)$. Show that

$$E[\Pi \mid Y = k] = \frac{a + b}{a + b + n} E[\Pi] + \frac{n}{a + b + n} \frac{k}{n}.$$

What happens as $n \rightarrow \infty$?

Solution

$E[\Pi] = \frac{a}{a+b}$ (from $\int_0^1 \pi \frac{\pi^{a-1}(1-\pi)^{b-1}}{B(a,b)} d\pi = \frac{B(a+1,b)}{B(a,b)} = \frac{a}{a+b}$). The posterior is $\text{Beta}(a+k, b+n-k)$, so

$$E[\Pi \mid k] = \frac{a+k}{a+b+n} = \frac{a+b}{a+b+n} \cdot \frac{a}{a+b} + \frac{n}{a+b+n} \cdot \frac{k}{n}.$$

As $n \rightarrow \infty$ the weight on the prior mean tends to 0 and the posterior mean tends to the sample proportion k/n (the MLE).